

A)

	Province	Gender	Age	Income	BMI	PhysicalActivity	Smoking	SelfRatedHealth	StressLevel
0	Ontario	female	59	80572.0	25.7	3.8	yes	Good	6.0
1	New Brunswick	Male	38	125739.0	24.2	3.4	No	Good	9.0
2	Nova Scotia	M	31	55997.0	35.7	NaN	No	Excellent	NaN
3	Saskatchewan	Male	30	75947.0	28.3	3.9	No	Excellent	10.0
4	Saskatchewan	female	79	113966.0	31.2	1.6	No	Fair	6.0

```
<class 'pandas.DataFrame'>
RangeIndex: 850 entries, 0 to 849
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Province              850 non-null   str
1   Gender                850 non-null   str
2   Age                  850 non-null   int64
3   Income               816 non-null   float64
4   BMI                  808 non-null   float64
5   PhysicalActivity      825 non-null   float64
6   Smoking              850 non-null   str
7   SelfRatedHealth      850 non-null   str
8   StressLevel          825 non-null   float64
dtypes: float64(4), int64(1), str(4)
memory usage: 59.9 KB
```

	Province	Gender	Age	Income	BMI	PhysicalActivity	Smoking	SelfRatedHealth	Sti
count	850	850	850.000000	816.000000	808.000000	825.000000	850	850	82
unique	10	6	NaN	NaN	NaN	NaN	6	5	
top	Quebec	Male	NaN	NaN	NaN	NaN	No	Good	
freq	100	230	NaN	NaN	NaN	NaN	552	258	
mean	NaN	NaN	49.517647	77461.917892	27.488861	3.511394	NaN	NaN	
std	NaN	NaN	18.192604	22497.833440	5.064344	1.925825	NaN	NaN	
min	NaN	NaN	18.000000	5000.000000	13.500000	0.000000	NaN	NaN	
25%	NaN	NaN	33.000000	63438.000000	24.000000	2.100000	NaN	NaN	
50%	NaN	NaN	49.000000	78130.000000	27.300000	3.500000	NaN	NaN	
75%	NaN	NaN	66.000000	90529.500000	30.900000	4.700000	NaN	NaN	
max	NaN	NaN	80.000000	250000.000000	60.000000	10.400000	NaN	NaN	1

Income is a quantitative variable because it represents a measurable numerical amount expressed in dollars. Each value has a meaningful magnitude, and it is valid to perform mathematical operations such as calculating the mean, median, and range. Since income has a true zero point and consistent differences between values, it fits a ratio scale and can be analyzed using standard statistical methods.

Province is a qualitative variable because it categorizes individuals into distinct groups based on geographic region. The values function as labels rather than numbers with mathematical meaning, and there is no inherent ordering or magnitude between provinces. As a result, analysis focuses on counts, proportions, or comparisons between categories rather than numerical calculations.

B)

```
Gender unique values: <StringArray>
['Female', 'Male']
Length: 2, dtype: str
Smoking unique values: <StringArray>
['Yes', 'No']
Length: 2, dtype: str
```

Missing values:

BMI 42

Income 34

StressLevel 25

PhysicalActivity 25

dtype: int64

JUSTIFICATION: Using MEDIAN IMPUTATION

- Median is resistant to outliers (important for Income and BMI)
- Preserves discrete nature of ordinal variables (StressLevel, PhysicalActivity)
- Missing values < 5% - deletion would reduce statistical power
- Missing pattern appears random (MAR)

BMI: Filled with median = 27.30

Income: Filled with median = 78130.00

StressLevel: Filled with median = 5.00

PhysicalActivity: Filled with median = 3.50

Missing values after imputation:

BMI 0

Income 0

StressLevel 0

PhysicalActivity 0

dtype: int64

Data types after conversion:

Age int64

Income float64

BMI float64

StressLevel int64

dtype: object

BMI_Category Distribution:

BMI_Category

Underweight 29

Normal 227

Overweight 343

Obese 251

Name: count, dtype: int64

IncomeBracket Distribution:

IncomeBracket

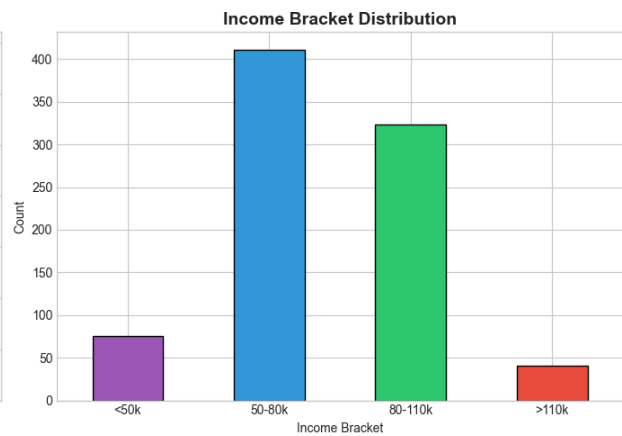
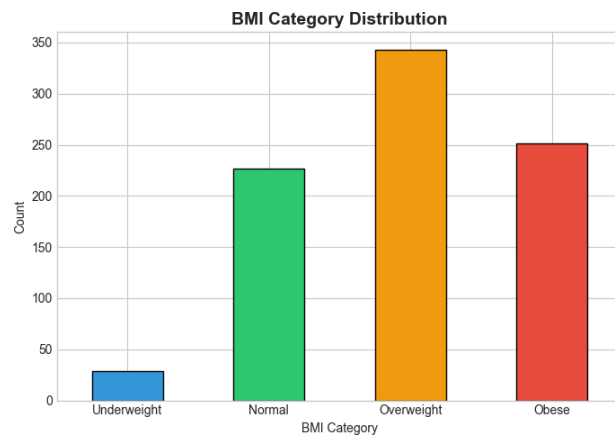
<50k 75

50-80k 411

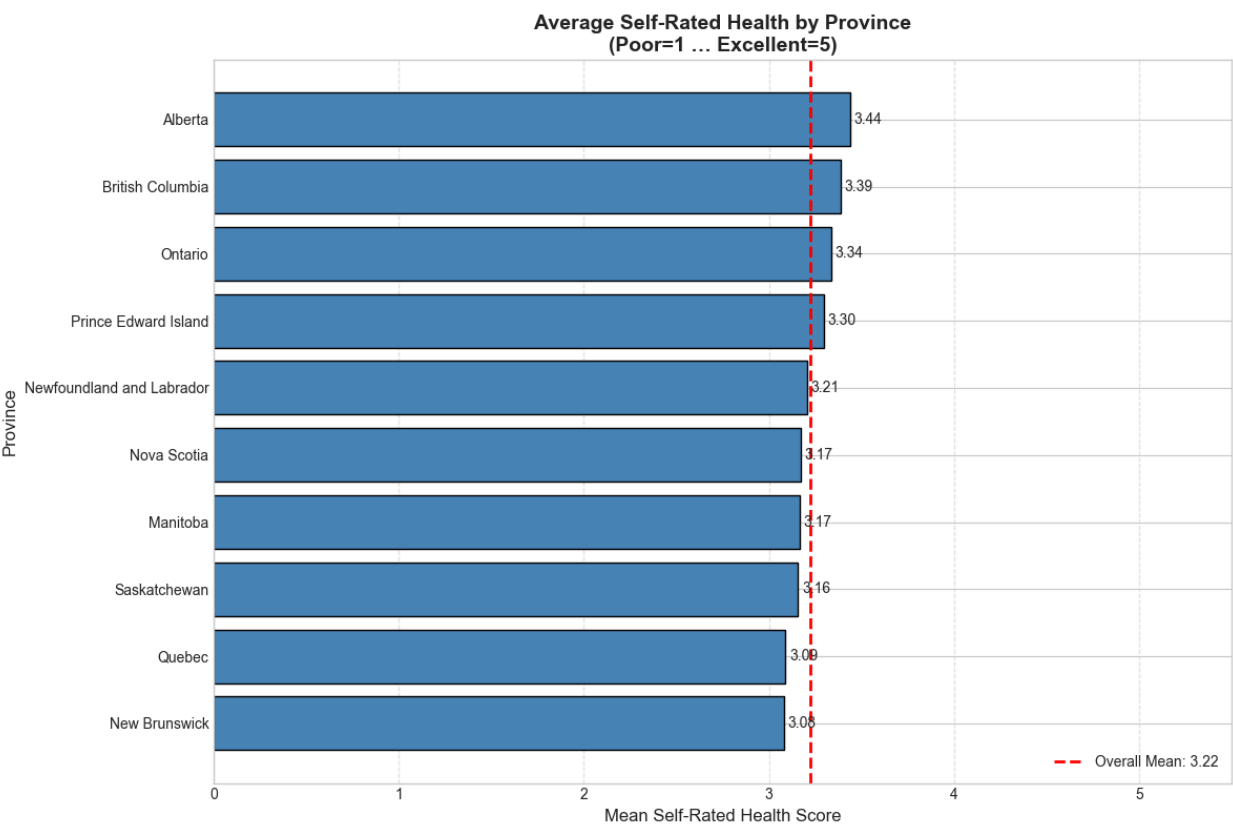
80-110k 323

>110k 41

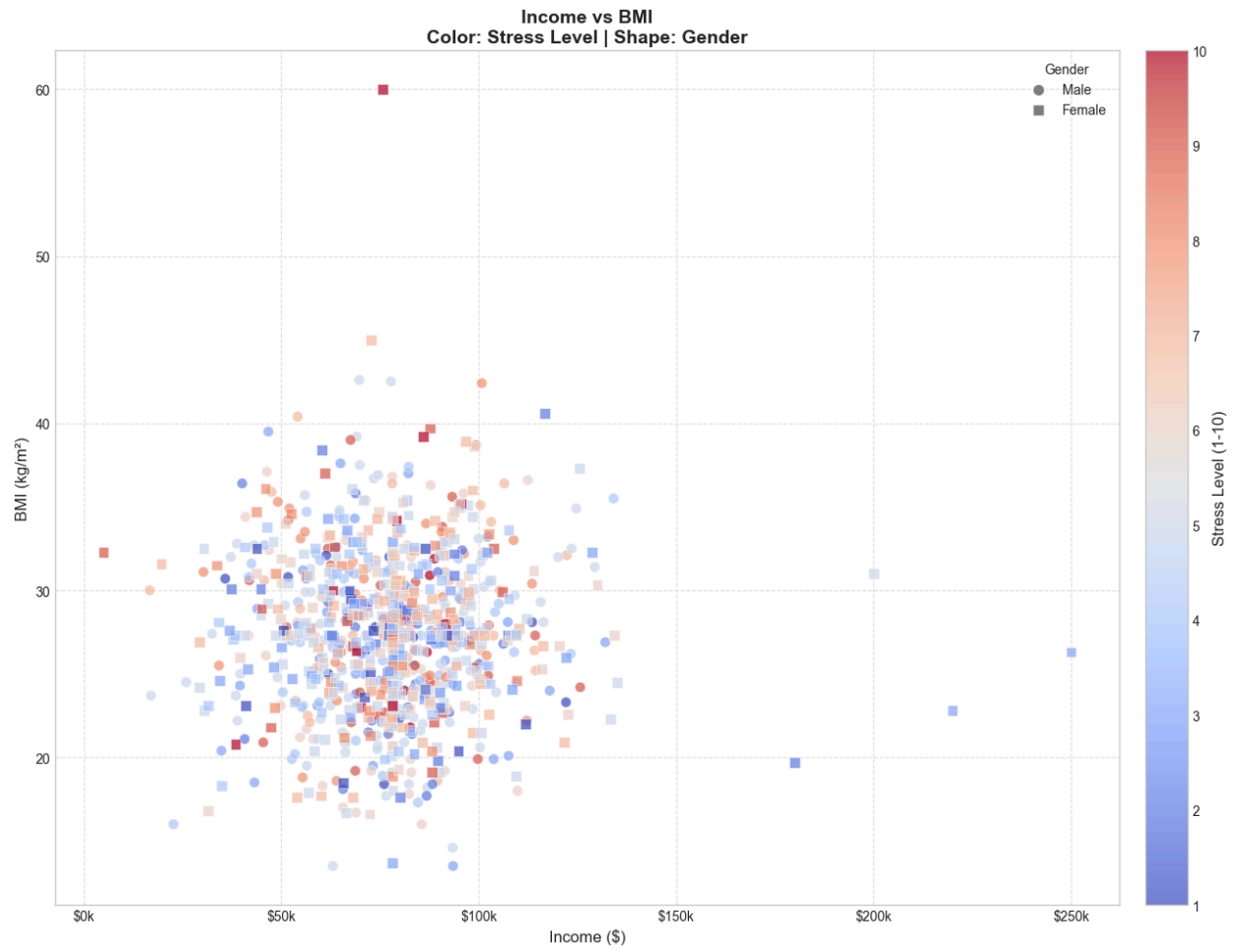
Name: count, dtype: int64



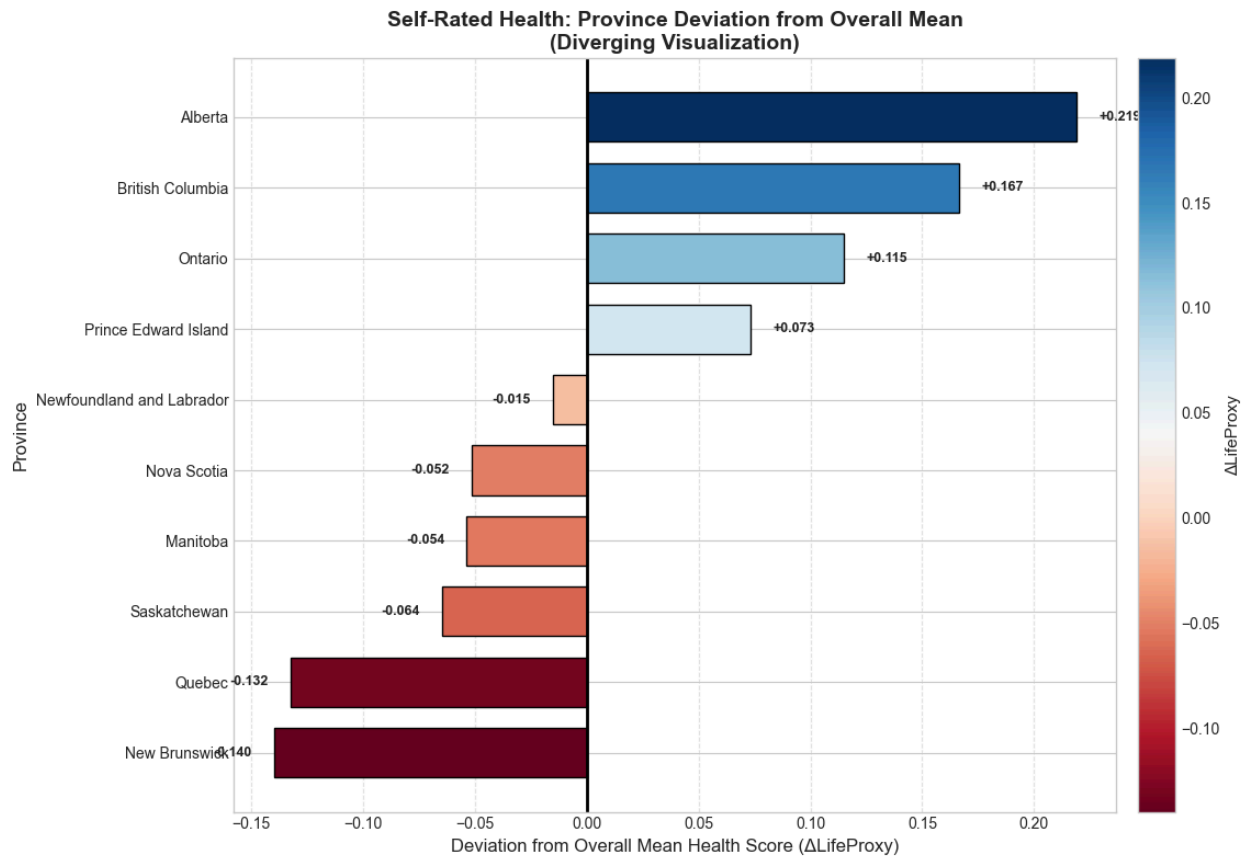
C1)



C2)



C3)



C4)

By employing suitable visual encodings and layout decisions, the visualizations were created to emphasize clarity and lessen cognitive load. Sequential palettes were used for ordered quantitative values to naturally reflect magnitude, color scales were deliberately chosen, and consistent, limited color use avoids superfluous visual noise. This eliminates the need for viewers to mentally decipher intricate legends in order to quickly interpret patterns.

By avoiding clutter, limiting the number of visual channels used simultaneously, and organizing components to facilitate simple comparison, cognitive load was reduced. In order to adhere to the contiguity principle, related information—such as labels, legends, and reference lines—was positioned close to the data it describes. Together, these design decisions guarantee that the images are simple to read, understand, and compare, allowing for the effective extraction of insights.

D1)

Since BMI values appear to be widely distributed across all income levels with no discernible upward or downward trend, the analysis reveals little to no strong relationship between income and BMI. This implies that income by itself is not a reliable indicator of body mass index in this

sample. While there are some differences between provinces, they are generally minor. Average health and BMI measurements vary slightly rather than significantly, indicating generally similar results across regions. The average BMI and self-reported health scores of males and females are essentially similar, and there is no discernible gender-specific pattern in the correlation between income and health metrics. Comparisons by gender also show negligible differences.

D2)

The existence of missing values in variables like income and BMI is one of the dataset's main limitations. If the missing values are not random, such as if respondents with lower incomes are less likely to report their income, handling this missingness through dropping or imputation may introduce bias. As a result, summary statistics may be impacted and observed relationships may be weakened, making the findings less indicative of actual population trends.